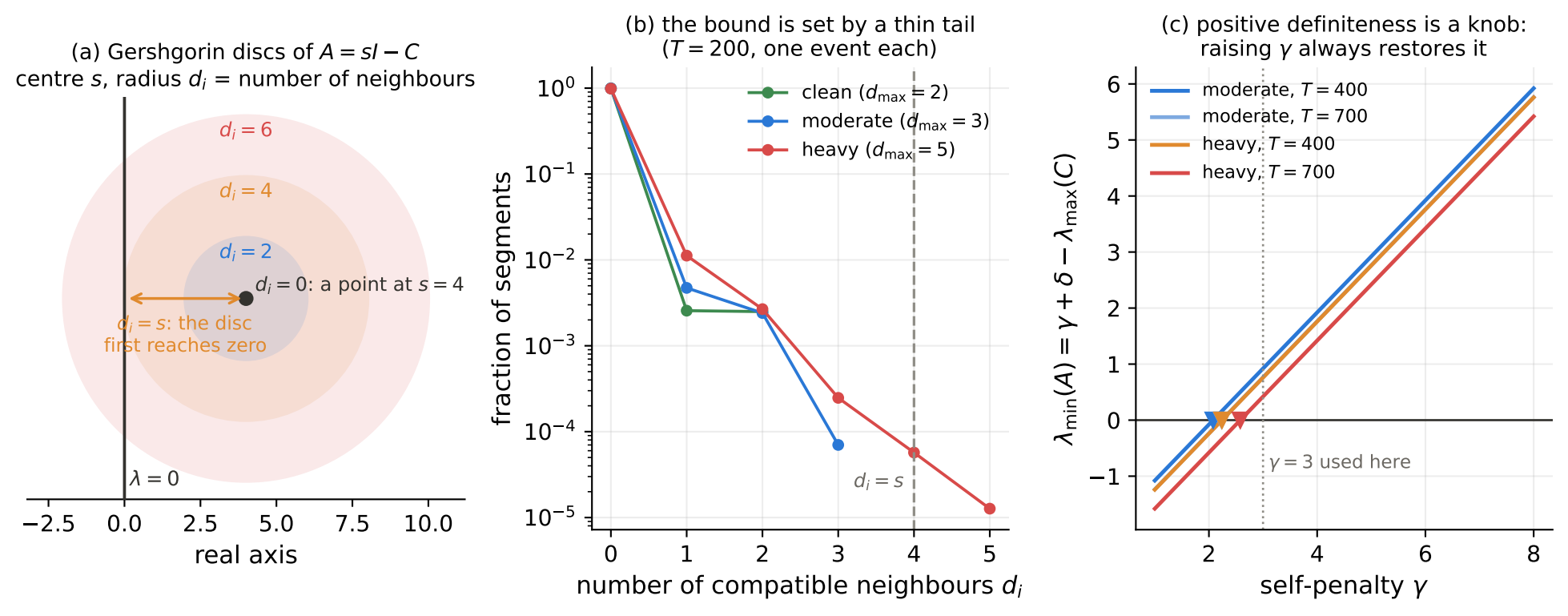
(a) Gershgorin discs of $A = sI - C$
centre s, radius d_i = number of neighbours
$d_i = 6$
$d_i = 4$
$d_i = 2$
$d_i = 0$: a point at $s = 4$
$d_i = s$: the disc
first reaches zero
$\lambda = 0$
−2.5
0.0
2.5
5.0
7.5
10.0
real axis
(b) the bound is set by a thin tail
($T = 200$, one event each)
clean ($d_{\max} = 2$)
moderate ($d_{\max} = 3$)
heavy ($d_{\max} = 5$)
fraction of segments
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
$d_i = s$
0
1
2
3
4
5
number of compatible neighbours d_i
(c) positive definiteness is a knob:
raising γ always restores it
moderate, $T = 400$
moderate, $T = 700$
heavy, $T = 400$
heavy, $T = 700$
$\lambda_{\min}(A) = \gamma + \delta - \lambda_{\max}(C)$
6
5
4
3
2
1
0
−1
$\gamma = 3$ used here
2
4
6
8
self-penalty γ